

“The Effect of Ambient Lighting on Cognitive Task Performance and Autonomic Activity During a Mental Arithmetic Task: An ECG-Based Within-Subject Study”

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Abstract

Ambient lighting is an important environmental factor that may influence both physiological responses and cognitive performance during mental demanding tasks. This study investigated the effect of bright and dim lighting on autonomic activity and cognitive task performance during a mental arithmetic task. Twenty participants completed both lighting conditions using a within-subject experimental design. Electrocardiogram (ECG) signals were recorded continuously using a Polar H10 chest-strap sensor. Heart rate (HR) and heart rate variability (RMSSD) were extracted from the ECG recordings, while cognitive performance was assessed using arithmetic task accuracy.

The normality of the difference scores was evaluated using the Shapiro–Wilk test. Paired-sample *t*-tests were used to compare HR and RMSSD between the two lighting conditions, whereas the Wilcoxon signed-rank test was applied to task accuracy due to non-normal data. Effect sizes and 95% confidence intervals were also calculated.

The statistical analyses showed no significant differences between bright and dim lighting for heart rate, RMSSD, or arithmetic task accuracy. Although small differences were observed, none reached statistical significance under the experimental conditions used in this study.

These findings suggest that ambient lighting alone did not produce measurable changes in autonomic activity or cognitive task performance during a short mental arithmetic task. Future studies with larger sample sizes, longer exposure periods, and additional physiological measures may provide a more comprehensive understanding of the relationship between environmental lighting and human performance.

1. Introduction

Ambient lighting is an important environmental factor that influences how people perform everyday cognitive activities. Students study in classrooms, employees work in offices, and individuals complete mentally demanding tasks under different lighting conditions for extended periods. Lighting has been shown to affect visual comfort, alertness, attention, and perceived mental workload. Appropriate lighting can improve comfort and reduce visual fatigue, whereas poor lighting conditions may negatively influence concentration and task performance. Consequently, understanding how lighting affects both physiological responses and cognitive performance has practical importance for the design of educational and workplace environments.

Physiological responses to cognitive workload can be assessed using electrocardiography (ECG), a non-invasive technique that records the electrical activity of the heart. ECG enables the measurement of heart rate (HR) and heart rate variability (HRV), both of which provide information about the activity of the autonomic nervous system. Heart rate generally reflects the body's immediate cardiovascular response, while HRV provides insight into the balance between sympathetic and parasympathetic nervous system activity. One of the most widely used time-domain HRV measures is the Root Mean Square of Successive Differences (RMSSD), which primarily reflects parasympathetic activity and short-term changes in autonomic regulation. Because HR and RMSSD are sensitive to mental workload and stress, they are frequently used in physiological computing research.

Mental arithmetic is commonly employed as an experimental task to induce cognitive workload because it requires continuous attention, working memory, and rapid mental processing. At the same time, task performance can be evaluated objectively using the number of problems attempted, the number of correct responses, and overall task accuracy. Combining physiological measurements with behavioural performance provide a more comprehensive understanding of how environmental factors influence human performance during cognitively demanding activities.

Although previous studies have investigated the effects of lighting on visual comfort, alertness, and productivity, relatively few studies have simultaneously examined physiological responses and cognitive performance using wearable ECG sensors in a controlled experimental setting. Furthermore, many existing studies have focused primarily on subjective measures of comfort rather than objective physiological indicators such as heart rate variability. Investigating both physiological and behavioural measures together provides a more complete assessment of the potential influence of ambient lighting.

The present study investigated the effect of bright and dim ambient lighting on heart rate, heart rate variability (RMSSD), and arithmetic task performance using a within-subject experimental design. Electrocardiogram (ECG) signals were continuously recorded using a Polar H10 chest-strap sensor while participants completed mental arithmetic tasks under both lighting conditions. To minimize individual differences, every participant completed both lighting conditions, and the presentation order was counterbalanced across participants.

The research question addressed in this study was:

How does ambient lighting influence heart rate variability and cognitive task performance during a mental arithmetic task?

Based on this research question, the following hypotheses were proposed:

- **H1:** Ambient lighting will significantly influence autonomic activity, reflected by changes in heart rate and heart rate variability (RMSSD).
- **H2:** Ambient lighting will significantly influence cognitive task performance, measured by arithmetic task accuracy.

By integrating wearable ECG measurements with behavioural performance, this study aims to contribute to physiological computing research and provide further evidence regarding the influence of ambient lighting on human physiology and cognitive performance.

2. Methodology

2.1.Participants

A total of 20 volunteers from Bauhaus-Universität Weimar participated in this study. All participants were healthy university students and completed both experimental lighting conditions. Participation was voluntary, and participants were informed about the purpose and procedure of the experiment before data collection began. To protect participant privacy, all ECG recordings and performance data were anonymised using participant identification numbers (P0011–P0033), and no personally identifiable information was stored.

2.2.Experimental Design

The study employed a within-subject experimental design to investigate the effect of ambient lighting on physiological responses and cognitive performance. The independent variable was the ambient lighting condition, which consisted of two levels: bright lighting and dim lighting. Each participant completed the mental arithmetic task under both lighting conditions, allowing each individual to serve as their own control and reducing variability caused by individual physiological differences.

The dependent variables included heart rate (HR), heart rate variability measured using the Root Mean Square of Successive Differences (RMSSD), and arithmetic task performance. Task performance was evaluated using the number of arithmetic problems attempted, the number of correct responses, and the overall task accuracy.

To minimise possible order effects, participants were divided into two equal groups. The first ten participants completed the experiment in the order **Bright** → **Dim**, while the remaining ten participants completed the conditions in the reverse order **Dim** → **Bright**. This **counterbalanced** design reduced potential learning and fatigue effects that could influence the experimental result.

2.3.Apparatus

Electrocardiogram (ECG) signals were recorded using a Polar H10 chest-strap sensor, which provides high-quality ECG recordings suitable for heart rate variability analysis. The sensor was positioned around the participant's chest according to the manufacturer's recommendations and continuously transmitted ECG data to a computer throughout the experiment.

A custom-developed web-based mental arithmetic application was used to present arithmetic problems during each task period. The application automatically recorded the number of attempted questions, the number of correct answers for each participant. The experiment was conducted in the indoor room for all participants, with only the ambient lighting level changed between the bright and dim conditions.

2.4.Experimental Procedure

Each participant completed both lighting conditions during a single experimental session lasting a little bit more than 14 minutes less than 15 minutes. Before the experiment began, the Polar H10 chest strap was attached, and the ECG signal quality was verified.

The session started with a 2-minute baseline period during which participants remained seated quietly to obtain a resting ECG recording. This baseline served as a physiological reference for the remainder of the experiment.

Following the baseline, participants completed a 5-minute mental arithmetic task under the first assigned lighting condition while ECG data were recorded continuously. After completing the first task, participants rested quietly for 2 minutes to allow partial physiological recovery. Finally, participants completed another 5-minute mental arithmetic task under the second lighting condition while ECG recording continued.

The arithmetic tasks presented problems of comparable difficulty under both lighting conditions to minimize learning effects. The order of the lighting conditions was counterbalanced across participants according to the assigned experimental group.

2.5.ECG Signal Processing

The recorded ECG signals were analyzed using Python together with the NeuroKit2 library. R-peaks were detected using the Pan–Tompkins algorithm, a widely used method for identifying ventricular depolarisation peaks in ECG recordings. The detected R-peaks were used to calculate consecutive RR intervals.

Heart rate (HR) was computed from the average RR intervals, while heart rate variability was quantified using the Root Mean Square of Successive Differences (RMSSD). HR and RMSSD were calculated separately for the baseline, bright lighting, recovery, and dim lighting segments for every participant. These participant-level results were then combined into a master dataset for subsequent statistical analysis.

2.6. Statistical Analysis

Statistical analyses were performed using Python with the SciPy library. Descriptive statistics were first calculated for all physiological and behavioural variables. Difference scores between the bright and dim lighting conditions were computed for heart rate, RMSSD, and arithmetic task accuracy.

The normality of the difference scores was assessed using the Shapiro–Wilk test. Since the assumptions of normality were satisfied for heart rate and RMSSD, paired-sample t-tests were used to compare these variables between the two lighting conditions. The difference scores for arithmetic task accuracy did not satisfy the normality assumption; therefore, a Wilcoxon signed-rank test was performed instead.

Effect sizes were calculated using Cohen's d for paired samples, and 95% confidence intervals were computed for the mean differences. Statistical significance was evaluated using a significance level of $\alpha = 0.05$.

3. Results

3.1. Descriptive Statistics

Twenty participants completed the experiment under both bright and dim lighting conditions are presented in Table 1.

	A	B	C	D	E	F	G	H	I	J
	master_statistics									
1	Metric	Mean Bright	Mean Dim	Mean Difference	95% CI Lower	95% CI Upper	Statistical Test	Statistic	p-value	Effect Size (Cohen's d)
2	Heart Rate	84.1	85.2	-1.1	-2.57	0.36	Paired t-test	-1.581	0.1303	-0.354
3	RMSSD	147.85	142.94	4.91	-12.42	22.23	Paired t-test	0.593	0.5604	0.133
4	Accuracy	82.47	84.98	-2.5	-9.13	4.12	Wilcoxon Signed-Rank	81.0	0.3884	-0.177

Table.1. Descriptive statistics for heart rate (HR), heart rate variability (RMSSD), and arithmetic task accuracy

The mean heart rate during the bright lighting condition was **84.10 bpm** (SD = 13.24), while the mean heart rate during the dim lighting condition was **85.20 bpm** (SD = 13.61). The mean difference between the two conditions was **-1.10 bpm**.

For heart rate variability, the mean RMSSD was **147.85 ms** (SD = 74.38) during the bright condition and **142.94 ms** (SD = 72.27) during the dim condition, resulting in a mean difference of **4.91 ms**.

Arithmetic task performance showed a mean accuracy of **82.47%** (SD = 12.50) under bright lighting

and **84.98%** (SD = 13.47) under dim lighting. The average difference between conditions was **-2.50%**.

Overall, the descriptive statistics indicate only small differences between the bright and dim lighting conditions across all measured variables.

3.2.Assumption Testing

Before conducting inferential statistical analyses, the normality of the difference scores was assessed using the Shapiro–Wilk test.

The difference scores for heart rate ($W = 0.992$, $p = .9996$) and RMSSD ($W = 0.941$, $p = .2472$) did not significantly deviate from normality. Therefore, paired-sample t-tests were considered appropriate for analysing these variables.

The difference scores for arithmetic task accuracy were not normally distributed ($W = 0.808$, $p = .0011$). Consequently, a non-parametric Wilcoxon signed-rank test was used to compare task accuracy between the two lighting conditions.

3.3.Inferential Statistics

A paired-sample t-test indicated that there was **no statistically significant difference** in heart rate between the bright and dim lighting conditions ($t(19) = -1.58$, $p = .130$, Cohen's $d = -0.35$). The mean difference was **-1.10 bpm**, with a 95% confidence interval ranging from **-2.57 to 0.36 bpm**.

Similarly, no statistically significant difference was observed for heart rate variability measured using RMSSD ($t(19) = 0.59$, $p = .560$, Cohen's $d = 0.13$). The mean difference between conditions was 4.91 ms, with a 95% confidence interval ranging from **-12.42 to 22.23 ms**.

Because the assumption of normality was violated for arithmetic task accuracy, a Wilcoxon signed-rank test was performed. The analysis showed **no statistically significant difference** between the bright and dim lighting conditions ($W = 81.0$, $p = .388$). The mean difference in task accuracy was **-2.50%**, with a 95% confidence interval ranging from **-9.13% to 4.13%**.

Overall, none of the statistical analyses identified significant differences between the bright and dim lighting conditions for heart rate, heart rate variability, or arithmetic task performance.

3.4.Data Visualization

The distributions of heart rate, RMSSD, and arithmetic task accuracy for the two lighting conditions are illustrated in **Figures 1–3** using boxplots. The figures show substantial overlap between the bright and dim lighting conditions, indicating similar distributions across participants.

Figures **4–6** present paired participant plots for heart rate, RMSSD, and arithmetic task accuracy. These plots demonstrate individual changes between the two lighting conditions and reveal noticeable inter-individual variability. While some participants exhibited higher values under bright lighting and others under dim lighting, no consistent trend favouring either lighting condition was observed.

The graphical analysis is consistent with the inferential statistical results, supporting the conclusion that ambient lighting produced only minor variations in physiological responses and arithmetic task performance under the conditions investigated in this study.

4. Discussion

This study investigated whether ambient lighting influences autonomic activity and cognitive performance during a mental arithmetic task. Heart rate (HR), heart rate variability (RMSSD), and arithmetic task accuracy were compared under bright and dim lighting conditions using a within-subject experimental design. Overall, the results indicated that ambient lighting did not produce statistically significant changes in any of the measured variables.

Although small differences were observed in the descriptive statistics, the inferential analyses showed that these differences were not statistically significant. Heart rate was slightly lower under bright lighting, while RMSSD was slightly higher. Task accuracy was marginally higher under dim lighting. However, all statistical tests produced p-values greater than 0.05, and the effect sizes were small, suggesting that the practical influence of lighting was limited under the conditions used in this study.

The graphical results support these findings. The boxplots showed substantial overlap between the bright and dim lighting conditions, while the paired participant plots revealed considerable individual variability. Some participants showed improvements under bright lighting, whereas others performed better under dim lighting. Because the responses varied across individuals, no consistent pattern emerged across the sample.

Several factors may explain these findings. The mental arithmetic task itself likely imposed a stronger cognitive demand than the lighting manipulation, making it difficult for relatively small changes in ambient lighting to produce measurable physiological effects. In addition, each lighting condition lasted only five minutes, which may not have been sufficient for lighting to meaningfully influence autonomic activity. Individual differences in factors such as stress, fatigue, fitness, breathing patterns, and baseline cardiovascular function may also have contributed to the variability observed in HR and RMSSD measurements.

Although no statistically significant effects were found, the study successfully demonstrated the complete physiological computing workflow. ECG data were collected using the Polar H10 sensor, processed with NeuroKit2 to extract HR and RMSSD, combined with behavioural performance measures, and analysed using appropriate statistical methods. This demonstrates how wearable physiological sensors can be integrated with cognitive experiments to evaluate human responses under controlled environmental conditions.

Future studies could increase the sample size, extend the duration of each lighting condition, and control additional factors such as sleep quality, caffeine intake, and respiration. Including other physiological measures, such as electrodermal activity (EDA) or eye-tracking data, may also provide a more comprehensive understanding of how environmental conditions influence cognitive performance and autonomic regulation.

Overall, the findings suggest that the ambient lighting conditions used in this experiment did not significantly affect heart rate, heart rate variability, or arithmetic task accuracy. Nevertheless, the study provides a useful framework for future physiological computing research involving environmental influences on human performance.

5. Limitations

This study has several limitations that should be considered when interpreting the findings. First, the sample size was relatively small ($n = 20$), which may have limited the statistical power to detect subtle differences between the lighting conditions. A larger sample would provide greater confidence in the findings and improve their generalisability.

Second, although every effort was made to maintain consistent experimental conditions, it was challenging to reproduce exactly the same dim lighting environment for every participant throughout the data collection period. Small variations in ambient illumination may have introduced additional variability into the physiological and behavioural measurements.

Furthermore, each lighting condition lasted only five minutes, which may not have been sufficient for ambient lighting to produce measurable changes in autonomic activity. Individual factors such as stress, fatigue, sleep quality, caffeine consumption, and breathing patterns were also not controlled and may have influenced heart rate and HRV measurements. Finally, this study focused on HR, RMSSD, and arithmetic task accuracy. Including additional physiological measures, such as electrodermal activity (EDA), respiration, or eye-tracking, could provide a more comprehensive understanding of the effects of ambient lighting on cognitive performance.

6. Conclusion

This study investigated the effect of ambient lighting on autonomic activity and cognitive performance during a mental arithmetic task. Heart rate (HR), heart rate variability (RMSSD), and arithmetic task accuracy were measured under both bright and dim lighting conditions using a within-subject experimental design.

The statistical analyses indicated that there were no significant differences between the two lighting conditions for any of the measured variables. Heart rate and RMSSD were analysed using paired-sample t-tests, while task accuracy was analysed using the Wilcoxon signed-rank test due to the non-normal distribution of the difference scores. All statistical tests produced p-values greater than the significance level of 0.05, and the observed effect sizes were small, suggesting that the influence of ambient lighting on physiological responses and cognitive performance was limited under the experimental conditions.

The graphical analyses supported these findings. The boxplots showed substantial overlap between the bright and dim lighting conditions, while the paired participant plots demonstrated considerable individual variability without a consistent trend favouring either lighting condition. These visual observations were consistent with the statistical results.

Although no statistically significant effects were observed, the project successfully demonstrated a complete physiological computing workflow. ECG signals were acquired using the Polar H10 sensor, processed in Python using NeuroKit2 to extract HR and RMSSD, combined with behavioural performance measures, and analysed using appropriate statistical methods. This workflow illustrates the practical application of wearable physiological sensing for evaluating human responses during controlled cognitive experiments.

Overall, the findings suggest that the ambient lighting conditions used in this study did not significantly affect heart rate, heart rate variability, or arithmetic task performance. Future research with larger sample sizes, longer exposure durations, more precisely controlled lighting conditions, and additional physiological measurements may provide a deeper understanding of how environmental factors influence cognitive performance and autonomic regulation.

7. References

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8. Tables

Fig: Statistical Comparison Between Bright and Dim Lighting Conditions

master_statistics									
Metric	Mean Bright	Mean Dim	Mean Difference	95% CI Lower	95% CI Upper	Statistical Test	Statistic	p-value	Effect Size (Cohen's d)
Heart Rate	84.1	85.2	-1.1	-2.57	0.36	Paired t-test	-1.581	0.1303	-0.354
RMSSD	147.85	142.94	4.91	-12.42	22.23	Paired t-test	0.593	0.5604	0.133
Accuracy	82.47	84.98	-2.5	-9.13	4.12	Wilcoxon Signed-Rank	81.0	0.3884	-0.177

9. Figures

A. Figure 1

Heart Rate under Bright and Dim Lighting Conditions

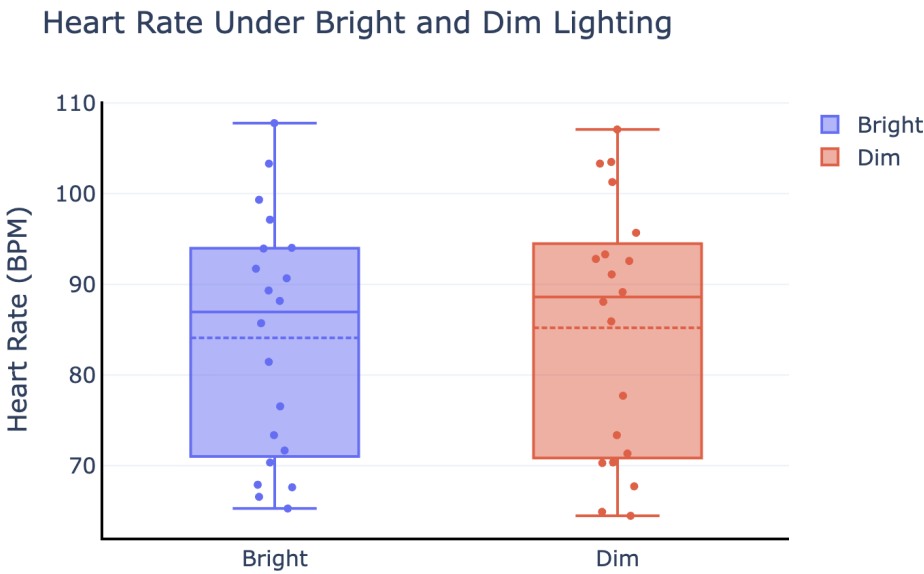


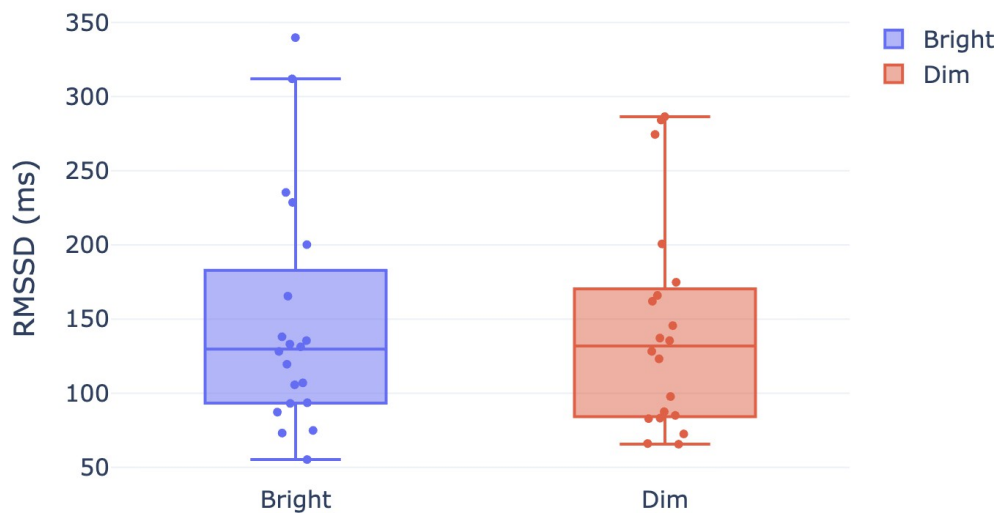
Figure 1. Boxplots showing the distribution of heart rate (HR) measured under bright and dim lighting conditions. Each point represents an individual participant.

B. Figure 2

Heart Rate Variability (RMSSD) under Bright and Dim Lighting Conditions

Figure 2. Boxplots illustrating the distribution of RMSSD values under bright and dim lighting conditions.

RMSSD Under Bright and Dim Lighting



C. Figure 3

Arithmetic Task Accuracy under Bright and Dim Lighting Conditions

Task Accuracy Under Bright and Dim Lighting

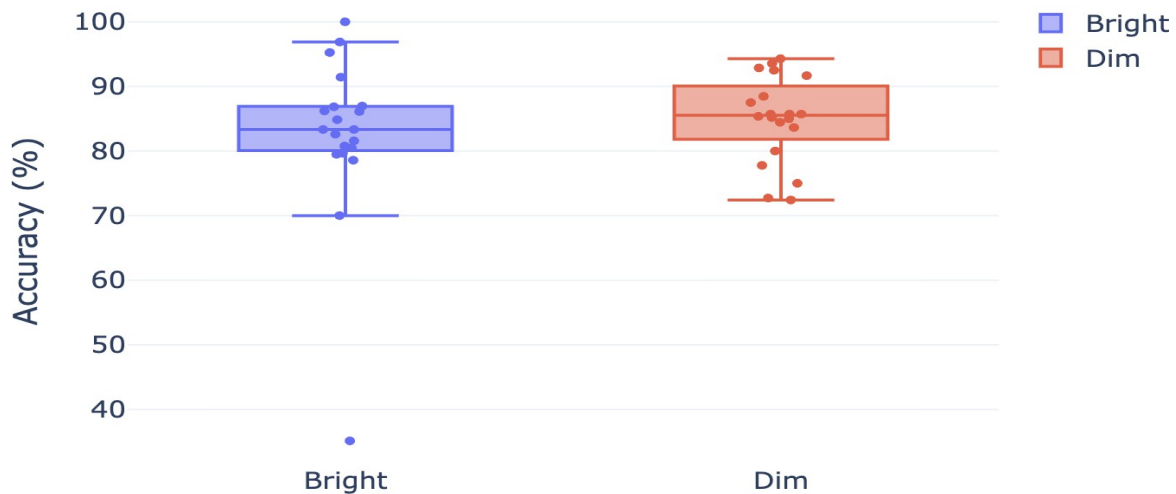


Figure 3. Boxplots showing arithmetic task accuracy for the bright and dim lighting conditions.

D. Figure 4

Individual Changes in Heart Rate

Individual Changes in Heart Rate

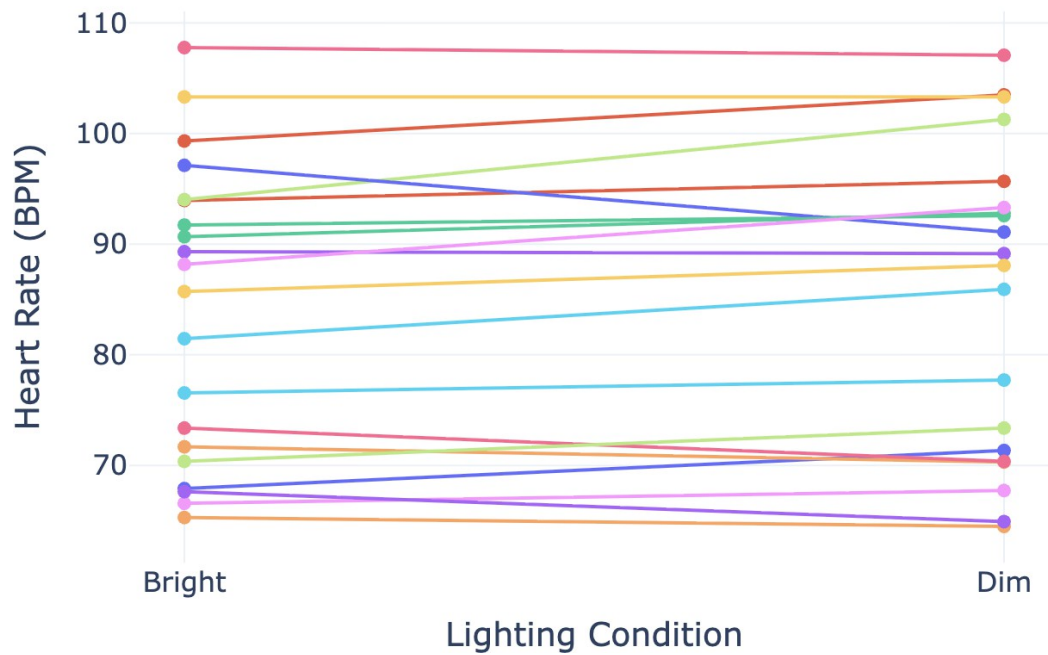


Figure 4. Paired participant plot illustrating individual heart rate changes between the bright and dim lighting conditions.

E. Figure 5

Individual Changes in RMSSD

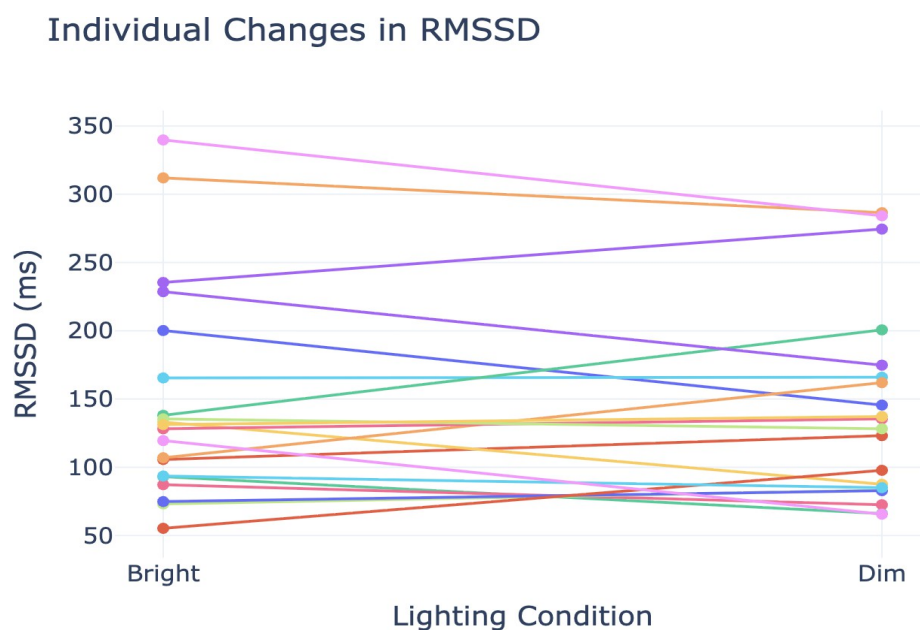


Figure 5. Paired participant plot showing changes in RMSSD for each participant across

the two lighting conditions.

F. Figure 6

Individual Changes in Task Accuracy.

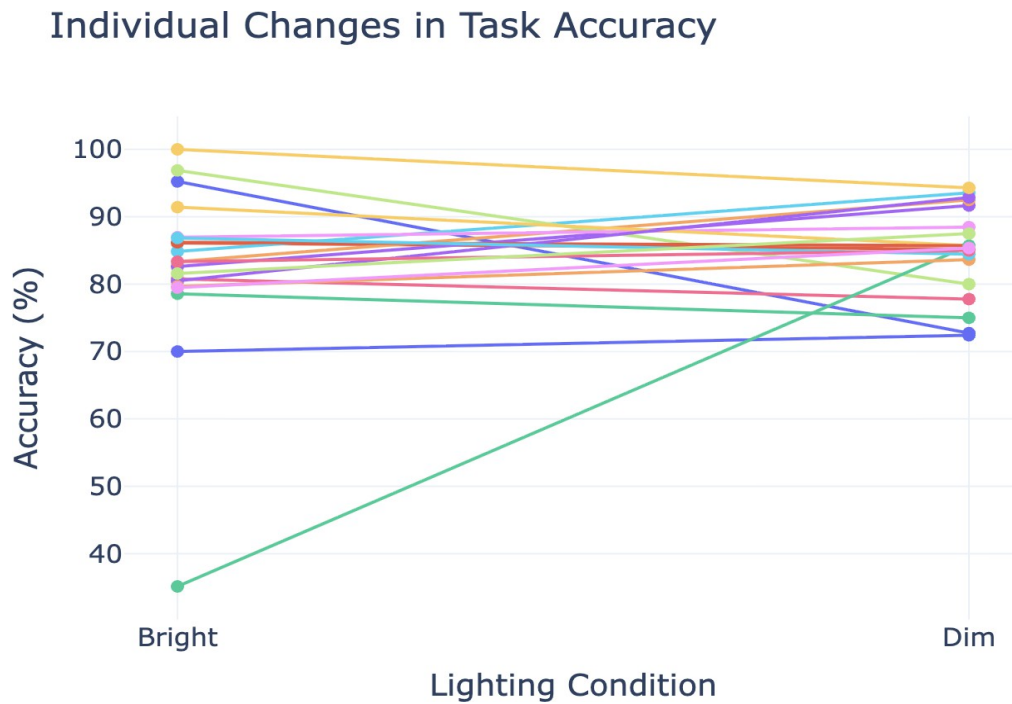


Figure 6. Paired participant plot illustrating changes in arithmetic task accuracy between the bright and dim lighting conditions for each participant.